

Term Project Proposal

DATS6401
Visualization of complex data
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This project analyzes a publicly available Online Shopping Dataset sourced from Kaggle. The dataset captures e-commerce transaction records. The primary objective is to uncover meaningful behavioral patterns among customers. Specifically examining how discount and coupon usage drives purchasing decisions, how online and offline spending differ across customer segments, and what early signals may indicate customer churn or disengagement

1. Feature List and Classification

The dataset contains 20 columns, classified into four types: Numerical, Categorical, Date, and ID. The table below summarizes each feature:

Feature	Type	Description
CustomerID	ID	Unique identifier per customer; excluded from analysis
Transaction_ID	ID	Unique identifier per transaction; excluded from analysis
Gender	Categorical	Customer gender (Male / Female)
Location	Categorical	Customer location or region
Product_SKU	Categorical	Stock Keeping Unit code for each product
Product_Description	Categorical	Text description of the product
Product_Category	Categorical	High-level product category
Coupon_Status	Categorical	Whether a coupon was used (Used / Not Used / Clicked)
Month	Categorical	Month of the transaction
Coupon_Code	Categorical	Code of the coupon applied, if any
Tenure_Months	Numerical	Months since customer first joined the platform
Quantity	Numerical	Number of items purchased per transaction

Avg_Price	Numerical	Average unit price of the product
Delivery_Charges	Numerical	Shipping fees for the transaction
GST	Numerical	Goods and Services Tax applied
Offline_Spend	Numerical	Total amount spent in offline channels
Online_Spend	Numerical	Total amount spent in online channels
Discount_pct	Numerical	Percentage discount applied to the transaction
Transaction_Date/Date	Date	Date of the transaction; used for time-series analysis

2. Feature Engineering

- Several features require engineering to maximize analytical value:
- Total_Spend: A new column will be derived by summing Online_Spend and Offline_Spend, providing a holistic view of each customer's total purchase value across both channels.
- Online_Ratio: Computed as $\text{Online_Spend} / \text{Total_Spend}$, this feature captures the proportion of a customer's activity that occurs online, enabling channel preference segmentation.
- Day_of_Week and Week_Number: Extracted from Transaction_Date to enable temporal analysis of purchasing behavior within and across weeks.
- Coupon_Used: A simplified binary flag derived from Coupon_Status (1 = coupon applied, 0 = not applied), making it easier to perform group comparisons and visualizations.

3. Planned Static Visualizations

- **Numerical Features**
 - Histograms: Tenure_Months, Avg_Price, Online_Spend, Offline_Spend, Total_Spend, and Discount_pct
 - Box Plots: Online_Spend and Offline_Spend grouped by Gender and Product_Category
 - Line Chart: Monthly trend of Total_Spend
 - Correlation Heatmap: across all numerical features to identify potential predictive relationships
- **Categorical Features**
 - Bar Charts: Gender, Product_Category, Location, and Month, Avg spend by Coupon_Status, Avg spend(Coupon_Status)
 - Stacked Bar Chart: Coupon & Discount Analysis (Coupon_Status by Category)

- Grouped Bar Chart: Online_Spend vs. Offline_Spend grouped by Location
- Scatter plot: Tenure vs Total_Spend